

Complete the table for each equation in 1-5.

Equation	Form	Key Feature(s) Revealed	Value(s) of Key Feature
1. $y = (x - 7)(x + 5)$	☐ Vertex form ☒ Factored form ☐ Standard form	x -intercepts Opening	(7,0) (-5,0) Up
2. $y = (x - 11)^2 - 23$	☒ Vertex form ☐ Factored form ☐ Standard form	Vertex Axis of symmetry Decrease Increase Opening	(11, -23) $x = 11$ ($-\infty$, 11) (11, ∞) Up
3. $y = -\frac{3}{8}x^2 - \frac{1}{4}x - 2$	☐ Vertex form ☐ Factored form ☒ Standard form	y -intercept Opening	(0, -2) Down
4. $y = -13(x + 12)^2$	☒ Vertex form ☐ Factored form ☐ Standard form	Vertex Opening Axis of symmetry Increase Decrease	(-12,0) Down $x = -12$ ($-\infty$, -12) (-12, $-\infty$)
5. $y = -x^2 + 11x - 52$	☐ Vertex form ☐ Factored form ☒ Standard form	y -int Opening	(0, -52) Down

6. Rewrite the quadratic equation in factored form.

$$y = x^2 + x - 56$$

$$y = (x + 8)(x - 7)$$

7. Rewrite the quadratic equation in standard form.

$$\begin{aligned}y &= (x - 9)^2 - 21 \\y &= x^2 - 18x + 81 - 21 \\y &= x^2 - 18x + 60\end{aligned}$$

8. Rewrite the equation in factored form.

$$\begin{aligned}y &= x^2 + 6x - 91 \\y &= (x + 13)(x - 7)\end{aligned}$$

9. Rewrite the equation in standard form.

$$\begin{aligned}y &= -7(x - 6)(x + 8) \\y &= -7(x^2 + 2x - 48) \\y &= -7x^2 - 14x + 336\end{aligned}$$

10. Laila is playing with a slingshot and shoots a ball straight up in the air. The ball is shot straight up from 2 meters above the ground with a velocity of 4 meters/second. The height of the ball is represented by the equation $y = -6x^2 + 4x + 2$, where x is time in seconds. Rewrite the equation in factored form.

$$\begin{aligned}y &= -2(3x^2 - 2x - 1) \\y &= -2(3x + 1)(x - 1)\end{aligned}$$